Jailbreaking Thesis Project

1. Selected References

Academic Literature

- **Tsmindashvili et al.** – *Improving LLM Outputs Against Jailbreak Attacks with Expert Model Integration.*
- **Saiem et al.** – *SequentialBreak: Large Language Models Can be Fooled by Embedding Jailbreak Prompts into Sequential Prompt Chains.*
- **Pernisi, F., Hovy, D., Rottger, P.** – *Compromesso! Italian Many-Shot Jailbreaks Undermine the Safety of Large Language Models.*
- **Joo, S., Koh, H., Jung, K.** – *Harmful Prompt Laundering: Jailbreaking LLMs with Abductive Styles and Symbolic Encoding.*
- **Yu, Z. et al.** – *Don't listen to me: Understanding and Exploring Jailbreak Prompts of Large Language Models.*

Non-Academic & Industry Sources

- **Cyberark Blog** – *Jailbreaking Every LLM With One Simple Click.*
- **SC Media** – *New LLM jailbreak method with 65% success rate developed by researchers.*
- **Anthropic Research** – *Many Shot Jailbreaking.*

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2. Problem Statement & Project Scope

Key Challenges (Aspects to focus on):

- **A. Data Obsolescence:** Jailbreaking techniques contained in older datasets may be too obsolete. There is a constant need to look for new datasets and evaluate against new benchmarks.
- **B. Shifting the Target:** Most violent/famous crimes are already secure from traditional techniques in most cases.
    - *Strategy:* Focus on **less traditional crimes** or **ethical concerns** (the "grey areas") which may produce harmful output from a moral standpoint, even if perfectly compliant and legal.

Future Research Question:

- **Prompt Injection vs. Jailbreaking:** Investigate whether there is a significant correlation between these two phenomena.

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3. Data Strategy & Resources

Primary Repositories:

- **JailBreakBench:** Contains data (JSON format) on successfully broken Llama-3 and GPT-4 models.
- **HarmBench Prompts:** Highly effective prompts that sometimes bypass even frontier models.
- **Multilingual Safety Benchmark (Xsafety project):** ~28,000 entries in different languages, useful to discover NLP patterns across languages.
- **Safety Prompts Project:** General repository for all data contained in the *SafetyPrompts* initiative.

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4. Experimental Contribution: A Predictive Risk-Scoring Framework

Objective: To transition the thesis from a purely descriptive NLP analysis to a proactive defense solution. The goal is to develop a lightweight, pre-filtering mechanism that assigns a **"Risk Probability Score"** to incoming prompts *before* they are processed by the target LLM.

Methodology: Leveraging the linguistic patterns identified in the analysis phase (specifically regarding "Ethical/Grey Area" attacks and "Prompt Injection"), the framework will function as a binary classifier (*Safe* vs. *Adversarial*) or a multi-class detector (*Safe*, *Financial Fraud*, *Injection*, *Ethical Bypass*).

4.1. Feature Engineering (The Inputs)

The model will not rely solely on keyword matching (which is easily bypassed) but on sophisticated NLP metrics derived from the selected datasets (JailbreakBench/HarmBench):

- **Syntactic Complexity:** Analyzing sentence structures, nesting levels, and dependency distances often used to obfuscate malicious intent.
- **Semantic Deviation:** Measuring the embedding distance (using BERT/RoBERTa) between the input prompt and known clusters of "refusal-inducing" prompts.

- **Injection Artifacts:** Specific detection of delimiter abuse (e.g., `###`, `---`), fake system commands, and context-switching patterns typical of Prompt Injection.

4.2. The Predictive Model

A comparative analysis of lightweight architectures to balance inference speed and detection accuracy:

- **Baseline:** Lightweight models (e.g., XGBoost, Random Forest).
- **Advanced:** Small Language Models (e.g., DistilBERT fine-tuned).

4.3. Validation Strategy

The framework will be tested on a **"held-out" set** of recent attacks (sourced from *JailbreakBench Artifacts* and *XSafety*).

- **Metric of Success: High Recall** (minimizing False Negatives) is prioritized over Precision.
    - *Rationale:* The cost of missing a successful jailbreak/injection in a production environment is significantly higher than flagging a false positive.

4.4. Expected Outcome

A proof-of-concept for an **"AI Firewall"** that uses the identified linguistic fingerprints to block **60-80%** of sophisticated, non-violent jailbreaks (financial/ethical) and prompt injections without needing to query the expensive and computationally heavy frontier model.